Python Bootcamp: Information

Welcome to the Python Bootcamp materials!

Here you will find all the information needed to study the *Python Bootcamp*, including **Lecture Slides**, and a set of **Exercise** materials (where answers will become available for all questions asked).

Our sessions will run on the following days:

Date	Day	Session time	Contents
12th January 2026	Monday	13:30 - 15:30	Introduction; First Steps with Python; Variables, Expressions & Statements
13th January 2026	Tuesday	10:00 - 12:00	Conditional Statements; An Introduction to Functions; Simple Repetition, Input & Output; Function Definition
		13:00 - 15:00	Iteration; Strings
14th January 2026	Wednesday	10:00 - 12:00	Lists; Dictionaries
		13:00 - 15:00	Global Variables; Tuples
15th January 2026	Thursday	10:00 - 12:00	Files; Advanced Features
		13:00 - 15:00	NumPy (for scientific computing)

16th January 2026	Friday	10:00 - 12:00	Pandas (data analysis & manipulation)
		13:00 - 15:00	Regular Expressions

They will be held in the **Abacws** Windows PC lab, **2.35**, which you will find on the second floor of the building. You may use the PCs there or your own laptops (recommended) for these sessions. We will be using *Google Colab* to run *Python* code, for which you will need a free *Google* account. If you do not have one, you may create one using *Google's* services.

The idea is that these sessions will allow us to *recap* or *learn the fundamentals of programming* for the first time, and they will provide you with the foundation needed to continue on to the more challenging and exciting aspects of your course. The materials will also remain available to you throughout the course, so you can come back to them whenever you choose to remind yourself of any aspect you wish, or to learn something in more detail which we only covered briefly in the sessions (after all, there is a *lot* to cover in a very short space of time!).

You can find all the lecture slides shown during the sessions in the **"Lecture Slides"** folder, and all exercise questions, files (and answers, when available) in the **"Exercises"** folder. Depending on the speed at which we work through the material together, we might not attempt all the listed exercises during the *Bootcamp*, but they will remain available to you in case you want to try them yourself, and test your knowledge. If you wish to discuss them with me, you're welcome to contact me about them at any time.

The learning outcomes of the *Python Bootcamp* (which is not assessed), are as follows. At the end of these sessions, you will be able to:

- Use the *Python* Interpreter, and write code in *Google Colab*;
- Understand and use arithmetic operators, assignment operators, statements and expressions;
- Realise the purpose of comments in code, and their benefit;
- Understand variable types, and how to pass variables to functions (by parameter, or as global objects);

- Predict the outcome of numeric expressions using the Order of Operations;
- Work with strings, arrays, maps and sets to add/remove/update/concatenate/retrieve elements;
- Understand how to use slicing to retrieve groups of elements;
- Know how to use built-in *Python* functions and methods (such as for splitting a string around a delimiter);
- Understand the meaning of mutable/immutable, and appreciate the challenges of nested arrays / loops;
- Use conditional statements and boolean logic to control the flow of program execution;
- Use for and while loops to repeat operations, and use Python keywords to end or continue them;
- Fetch and process input from the user;
- Format and display information to the user;
- Create and use functions, and lambda functions;
- Explain the general purpose of Objects, and that they have Methods and Attributes (properties);
- Explain the different types of programming error (syntax, runtime and semantic);
- Use built-in Python functions, standard modules (e.g. math), and third-party modules (such as NumPy and Pandas);
- Know how to open, read and write to files, on your local machine or in *Colab's* temporary storage;
- Understand the basics of writing code to implement algorithms;
- Use *NumPy* to process data arrays efficiently;
- Use *Pandas* to read, segment and process tabular data;
- Use *Regular Expressions* to detect and isolate specific patterns in textual data.

I hope you will enjoy the *Bootcamp* sessions, and please feel free to raise any questions or suggestions with me, at any time. I can be reached:

- In person (I will be leading all sessions);
- Via e-mail at HumphreysDJ@cardiff.ac.uk (mailto:HumphreysDJ@cardiff.ac.uk);
- Via *Teams* (you are welcome to message me directly).

I look forward to seeing you!

Best wishes,

Dr. David J. Humphreys